

AS Physics & Mathematics Mechanics 1

Homeschool Teaching Plan

10-month online course · 1.5-hour sessions · 2x/week per subject

Target sitting: Summer (May/June) 2027

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1. Course Overview

This plan covers the two International Advanced Subsidiary (IAS) units of Edexcel International A Level Physics (Units 1, 2 and 3) and the Mechanics 1 (M1) unit of Edexcel International A Level Mathematics, delivered one-to-one online in 1.5-hour sessions, two sessions per week per subject (4 sessions/week total), over roughly 43 weeks from mid-July 2026 to early May 2027, ending with the Summer 2027 examination series.

2. Physics - Unit Summary

Note: at least 40% of the marks across all Physics papers target mathematics skills at Level 2 or above (graphs, trigonometry, algebraic rearrangement, standard form) - this is reinforced throughout by the parallel Maths M1 sessions.

3. Mathematics M1 - Unit Summary

4. Phase-by-Phase Teaching Schedule

The schedule below assumes a start date of Monday 13 July 2026; shift all dates equally if the actual start date differs. Each week = 2 Physics sessions + 2 Maths sessions (1.5 h each).

Phase 0 - Orientation & Diagnostic (Week 1)

Phase 1 - Physics Unit 1: Mechanics | Maths M1: Models, Kinematics, Dynamics (Weeks 2-13)

Phase 2 - Physics Unit 1: Materials | Maths M1: Statics, Moments (Weeks 14-19)

Mock Checkpoint 1 & Review (Weeks 20-21)

Holiday Buffer (Weeks 22-25)

Phase 4 - Physics Unit 2: Waves & Particle Nature of Light | Maths M1: Vectors + Past-Paper Practice (Weeks 26-37)

Phase 5 - Physics Unit 2: Electric Circuits + Unit 3 Consolidation | Maths M1: Past-Paper Practice (Weeks 38-40)

Mock Checkpoint 2 (Week 41)

Final Revision (Weeks 42-43)

Exam window: the Summer 2027 International A Level series is provisionally scheduled to begin around 4 May 2027. Confirm the exact WPH11, WPH12, WPH13 and WME01 dates/times on Pearson's final Summer 2027 timetable once published, and adjust the final 2-3 weeks of revision accordingly.

5. Physics Core Practicals Checklist (Units 1 & 2)

Since sessions are online, practicals can be delivered via video demonstration + data analysis, remote-lab simulations (e.g. PhET, Multisim, IsaacPhysics), or by asking the student to run a simplified version at home under guidance - Unit 3 (WPH13) tests understanding of method, uncertainty and analysis rather than requiring lab attendance.

6. Recommended Resources

Textbooks: Pearson Edexcel International A Level Physics Student Book 1 (Units 1-3); Pearson Edexcel International A Level Mathematics Mechanics 1 Student Book

Official specifications & past papers: [qualifications.pearson.com](https://www.pearson.com/qualifications) (search “International Advanced Level Physics” / “Mathematics” specifications and past papers)

Past papers & mark schemes: [savemyexams.com](https://www.savemyexams.com), [physicsandmathstutor.com](https://www.physicsandmathstutor.com) (PMT) - both index Edexcel IAL Physics and M1 papers by topic

Formula sheet & data sheet: provided by Pearson in the exam - practise using it from the start so the student is fluent by the mocks

Practical/virtual lab support: PhET Interactive Simulations, [IsaacPhysics.org](https://www.isaacphysics.org) for extra problem sets and practical-skills style questions

7. Planning Notes & Flexibility

This is a working scaffold, not a rigid script - use the Week 1 diagnostic and ongoing homework performance to compress or stretch individual topics.

Because the student's starting level hasn't been assessed yet, Weeks 2-5 build in natural checkpoints (kinematics, then dynamics) where pace can be adjusted before committing to the rest of the schedule.

The plan assumes M1 is being taught as a standalone applied unit; if Pure Mathematics 1 (P1) also needs to be covered for a full AS Maths award, a separate P1 track (or additional weekly sessions) will need to be added.

Two full mock checkpoints (Weeks 20 and 41) are built in deliberately - timed, full-paper conditions matter as much as content coverage for IAL-style exams.

Holiday buffer (Weeks 22-25) can absorb schedule slippage; if the course is running ahead, use it for extra past-paper practice instead.

Item	Detail
Awarding body	Pearson Edexcel - International Advanced Level (IAL)
Physics units covered	Unit 1: Mechanics and Materials (WPH11) · Unit 2: Waves and Electricity (WPH12) · Unit 3: Practical Skills in Physics I (WPH13)
Maths unit covered	Mechanics 1 / M1 (WME01) - the applied unit only; Pure Mathematics 1 is assumed to be studied separately
Session structure	2 × 1.5 h/week for Physics + 2 × 1.5 h/week for Maths M1 (4 sessions/week, ~6 h/week total)
Course length	~43 teaching weeks (13 Jul 2026 → early May 2027), including 2 built-in mock checkpoints and a holiday buffer
Target exam series	Summer 2027 (begins ~4 May 2027, provisional - confirm exact WPH11/WPH12/WPH13/WME01 dates on Pearson's final Summer 2027 timetable nearer the time)
Starting point	Student's IGCSE-level foundation has not yet been formally assessed - Week 1 includes a diagnostic test in both subjects to calibrate pacing

Unit	Content Focus	Assessment	Weighting
Unit 1 WPH11	Mechanics (kinematics, forces, momentum, moments, work/energy/power) + Materials (density, viscosity, Hooke's Law, Young modulus)	Written exam, 1 h 30, 80 marks. Multiple-choice, short/open-response, calculation & extended-writing questions.	40% of IAS / 20% of full IAL
Unit 2 WPH12	Waves and Particle Nature of Light (wave properties, standing waves, refraction, diffraction, photoelectric effect, de Broglie) + Electric Circuits (current, resistance, potential dividers, e.m.f.)	Written exam, 1 h 30, 80 marks. Same question styles as Unit 1.	40% of IAS / 20% of full IAL
Unit 3 WPH13	Practical Skills in Physics I - tests knowledge of experimental procedures/techniques developed across the 8 core practicals in Units 1 & 2, plus data analysis and uncertainty	Written exam, 1 h 20, 50 marks.	20% of IAS / 10% of full IAL

Unit	Content Focus (6 chapters)	Assessment
M1 WME01	1. Mathematical Models in Mechanics 2. Kinematics of a Particle Moving in a Straight Line 3. Dynamics of a Particle Moving in a Straight Line or Plane 4. Statics of a Particle 5. Moments 6. Vectors in Mechanics	Written exam, 1 h 30, availability January/June/October. Standard IAL Mathematics format.

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 1	13 Jul 2026	Orientation; diagnostic test covering IGCSE foundations (units & SI prefixes, graphs, basic vectors, equations rearrangement)	Diagnostic test covering prerequisite maths (algebra, indices, surds, trigonometry, straight-line graphs); calculator & formula-booklet familiarisation

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 2	20 Jul 2026	SUVAT equations; displacement-time, velocity-time and acceleration-time graphs	Ch1 Mathematical Models in Mechanics - modelling assumptions (particle, rod, smooth, light, inextensible, etc.)
Wk 3	27 Jul 2026	Scalars vs vectors; resolving vectors; resultant of coplanar vectors	Ch2 Kinematics - constant-acceleration formulae (SUVAT), motion graphs
Wk 4	03 Aug 2026	Projectile motion - independence of horizontal/vertical components	Ch2 Kinematics - variable acceleration using calculus (differentiating/integrating s, v, a)
Wk 5	10 Aug 2026	Free-body diagrams; Newton's 1st & 2nd Law ($\Sigma F = ma$); terminal velocity	Ch2 consolidation - mixed kinematics problems, exam-style questions
Wk 6	17 Aug 2026	Weight & gravitational field strength; CORE PRACTICAL 1 - acceleration of a freely-falling object	Ch3 Dynamics - Newton's laws, force diagrams, resolving forces on an incline
Wk 7	24 Aug 2026	Newton's 3rd Law; momentum; conservation of linear momentum	Ch3 Dynamics - connected particles: strings & pulleys
Wk 8	31 Aug 2026	Moments; centre of gravity; equilibrium of an extended rigid body	Ch3 Dynamics - friction; motion on inclined planes
Wk 9	07 Sep 2026	Work done (incl. force not along the line of motion); kinetic energy	Ch3 consolidation - mixed dynamics problems, exam-style questions
Wk 10	14 Sep 2026	Gravitational potential energy; conservation of energy	Mixed kinematics + dynamics problem set (past-paper style)
Wk 11	21 Sep 2026	Power; efficiency (energy and power forms)	Problem-solving clinic - targeted at weak areas from homework
Wk 12	28 Sep 2026	Unit 1 Mechanics consolidation - mixed past-paper questions	Ch1-3 timed past-paper practice
Wk 13	05 Oct 2026	Unit 1 Mechanics past-paper practice / buffer catch-up	Buffer / catch-up week

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 14	12 Oct 2026	Density; upthrust (Archimedes' principle)	Ch4 Statics of a Particle - equilibrium under coplanar forces
Wk 15	19 Oct 2026	Viscosity, Stokes' Law; CORE PRACTICAL 2 - falling-ball method for viscosity of a liquid	Ch4 Statics - friction in equilibrium problems
Wk 16	26 Oct 2026	Hooke's Law; stress, strain, Young modulus	Ch5 Moments - moment of a force, rigid-body equilibrium
Wk 17	02 Nov 2026	CORE PRACTICAL 3 - Young modulus; force-extension graphs; elastic/plastic deformation, yield point	Ch5 Moments - ladders, beams, more complex problems
Wk 18	09 Nov 2026	Stress-strain graphs; breaking stress; elastic strain energy	Ch4-5 consolidation - mixed statics/moments problems
Wk 19	16 Nov 2026	Unit 1 (Mechanics + Materials) full past-paper practice (WPH11 style)	M1 Ch1-5 combined timed past-paper practice (mock prep)

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 20	23 Nov 2026	MOCK 1 - WPH11-style full timed paper (1 h 30, 80 marks); mark & review	MOCK 1 - M1 mixed paper covering Ch1-5; mark & review
Wk 21	30 Nov 2026	Review week - targeted remediation on weak topics from Mock 1	Review week - targeted remediation on weak topics from Mock 1

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 22-24	07-21 Dec 2026	Flexible / holiday buffer - optional light sessions, catch-up on any topic behind schedule, or full rest	Flexible / holiday buffer - same as Physics; adjust around family holiday plans
Wk 25	28 Dec 2026	Holiday - no formal sessions	Holiday - no formal sessions

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 26	04 Jan 2027	Wave basics - amplitude, frequency, period, $v = f\lambda$; transverse vs longitudinal waves	Ch6 Vectors - position vectors, i-j notation, vector methods for displacement/velocity
Wk 27	11 Jan 2027	Standing/stationary waves - formation, nodes & antinodes; CORE PRACTICAL 4 - speed of sound (oscilloscope)	Ch6 Vectors - relative motion, closest-approach problems
Wk 28	18 Jan 2027	Superposition, coherence, path difference, phase; wave speed on a string; CORE PRACTICAL 5 - vibrating string	Ch6 consolidation - mixed vector problems
Wk 29	25 Jan 2027	Intensity of radiation; refraction, Snell's Law, refractive index	Full M1 past paper 1 (timed) + review
Wk 30	01 Feb 2027	Critical angle, total internal reflection; measuring refractive index of a solid	Full M1 past paper 2 (timed) + review
Wk 31	08 Feb 2027	Polarisation; diffraction, Huygens' construction, diffraction-grating equation; CORE PRACTICAL 6 - wavelength via diffraction grating	Topic clinic - weakest areas from running error log
Wk 32	15 Feb 2027	Wave-particle duality - electron diffraction, de Broglie equation	Full M1 past paper 3 (timed) + review
Wk 33	22 Feb 2027	Photoelectric effect - photon model, $E = hf$, threshold frequency, work function	Topic clinic
Wk 34	01 Mar 2027	Photoelectric equation, electronvolt; atomic line spectra & energy-level transitions	Full M1 past paper 4 (timed) + review
Wk 35	08 Mar 2027	Pulse-echo techniques; wave vs photon model summary; Unit 2 Waves consolidation	Topic clinic / exam technique - command words, mark-scheme language
Wk 36	15 Mar 2027	Unit 2 Waves - full past-paper practice (WPH12-style, Waves section)	Full M1 past paper 5 (timed) + review
Wk 37	22 Mar 2027	Buffer / consolidation - Waves and Particle Nature of Light	Consolidation - formula-sheet mastery, buffer

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 38	29 Mar 2027	Current, charge, p.d., resistance, Ohm's Law; series & parallel resistance derivations	Full M1 past paper 6 (timed) + review
Wk 39	05 Apr 2027	Power equations, resistivity; CORE PRACTICAL 7 - resistivity of a material; I-V graphs (ohmic, filament, thermistor, diode)	Full M1 past paper 7 (timed) + review
Wk 40	12 Apr 2027	Potential dividers; e.m.f. & internal resistance; CORE PRACTICAL 8; thermistors/LDRs; Unit 3 Practical Skills consolidation (all 8 CPs + uncertainty analysis)	Final formula & technique consolidation

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 41	19 Apr 2027	MOCK 2 - full timed mock set: WPH11, WPH12, WPH13-style papers; detailed mark & review	MOCK 2 - full timed M1 mock paper; detailed mark & review

Week	Dates	Physics (Edexcel IAL)	Mathematics M1 (Edexcel IAL)
Wk 42-43	26 Apr - 03 May 2027	Final revision - targeted gap-filling from Mock 2, formula-sheet drilling, topic-by-topic past-paper questions, exam-day timing strategy	Final revision - same approach; past-paper questions by topic, exam-day timing strategy

CP #	Practical
CP 1	Determine the acceleration of a freely-falling object
CP 2	Use a falling-ball method to determine the viscosity of a liquid
CP 3	Determine the Young modulus of a material
CP 4	Determine the speed of sound in air using a 2-beam oscilloscope, signal generator, speaker and microphone
CP 5	Investigate the effects of length, tension and mass per unit length on the frequency of a vibrating string or wire
CP 6	Determine the wavelength of light from a laser or other light source using a diffraction grating
CP 7	Determine the electrical resistivity of a material
CP 8	Determine the e.m.f. and internal resistance of an electrical cell